VARIATIONAL IMAGE RESTORATION: DECONVOLUTION, POISSON EDITING, AND STRUCTURE-GUIDED INPAINTING

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November 2023

Abstract. Fundamental restoration tasks in computational photography—including non-blind image deblurring, seamless region cloning, and digital inpainting—can be unified as variational optimization problems on a discrete two-dimensional pixel lattice. In this paper, we develop the mathematical theory and matrix-free numerical solvers for three closely linked problems: (i) non-blind deconvolution via Steepest Descent, isotropic Tikhonov–Laplacian Conjugate Gradient (CG), and structure-guided anisotropic CG optimization; (ii) gradient-domain Poisson and structure-dominant mixed-gradient seamless cloning; and (iii) digital image inpainting via harmonic PDE relaxation, Telea's Eikonal Fast Marching Method (FMM), and a novel **FMM-Warm-Started Structure-Tensor Anisotropic CG Inpainting** scheme. We prove that both the isotropic five-point Laplacian $L = G^T G$ and the structure-tensor-weighted divergence operator $L_D = G^T D G$ are symmetric positive definite under Dirichlet boundary conditions, establish that FMM warm-starting strictly contracts the Krylov initial energy norm $\|X_0 - X^*\|_{L_D}$, and present clean matrix-free algorithms for each solver.

1. Introduction

Photographs captured in real-world environments rarely record an ideal, uncorrupted scene. Optical defocus and camera motion blur smear fine spatial detail across neighboring pixels; sensor noise corrupts low-light exposures; and physical obstructions, surface scratches, or unwanted foreground objects occlude portions of the scene. A particularly compelling modern application of digital inpainting is automated privacy and copyright protection: a camera system can detect restricted landmarks or fiducial QR markers in a live scene, generate a binary mask over the protected region, and seamlessly reconstruct the background across the mask so the restricted object never appears in the final photograph. Similarly, archival restoration and reflection removal rely on recovering sharp, visually coherent structure from degraded pixel observations.

At first glance, deblurring a motion-blurred photograph, seamlessly pasting an object from one scene into another, and filling in a missing hole appear to be three separate problems requiring unrelated algorithms. Yet when we formulate each task mathematically on a discrete pixel grid $\Lambda = \{0, \dots, W-1\} \times \{0, \dots, H-1\}$, a striking unification emerges. In each case, we seek an unknown image X that minimizes a convex quadratic energy functional measuring discrepancy in either pixel intensities or spatial gradients. Setting the first derivative of this energy functional to zero reduces all three problems to solving a large-scale symmetric positive definite (SPD) linear system $AX = b$.

The central computational challenge is dimensionality. Even a modest 500×500 image contains $N = 250{,}000$ pixels per channel, meaning the explicit matrix $A \in \mathbb{R}^{N \times N}$ would contain 6.25×10^{10} entries—far too large to store in memory or invert via Gaussian elimination ($\mathcal{O}(N^3)$). Fortunately, every linear operator arising in convolution and discrete differential geometry is sparse and shift-invariant: we can evaluate the matrix-vector product $v \mapsto Av$ directly on the two-dimensional image grid using small local convolution stencils in $\mathcal{O}(N)$ time without ever forming the matrix A. This observation makes iterative Krylov subspace solvers—and specifically the Conjugate Gradient (CG) method—the ideal mathematical engine for variational image restoration.

Alongside variational PDE solvers, we also study geometric front-propagation inpainting via the Fast Marching Method (FMM) introduced by Telea [2]. While harmonic Poisson inpainting produces smooth membrane interpolants globally, it requires $\mathcal{O}(\mathrm{diam}(\Omega))$ CG iterations for boundary information to diffuse from $\partial\Omega$ to the center of the missing domain Ω, and its isotropic penalty blurs sharp edges that cross the mask. Conversely, Telea's FMM propagates an Eikonal arrival-time front $|\nabla T(x,y)| = 1$ inward in a single $\mathcal{O}(|\Omega| \log |\Omega|)$ pass, but because it never relaxes globally, colliding wave fronts leave visible medial-axis ridge artifacts near the center of large holes.

To overcome the limitations of both classical approaches, we introduce three structure-guided algorithmic innovations across our variational pipeline: (1) **Structure-Guided Anisotropic CG Deconvolution** ($\lambda L_D = \lambda G^T D G$), which replaces isotropic Laplacian regularization with a structure-tensor conductivity field so noise is smoothed in flat regions without blurring real image edges; (2) **Structure-Dominant Mixed-Gradient Poisson Cloning**, which preserves salient background textures across transparent or concave foreground masks; and (3) **FMM-Warm-Started Structure-Tensor Anisotropic CG Inpainting**, which uses Telea's non-iterative FMM output

X_{FMM} as a warm-start initial guess X_0 for Conjugate Gradient on $M_\Omega \odot (G^T D G X) = 0$, eliminating FMM's medial-axis seams, preserving sharp cross-boundary edges, and reaching $> 40.1\ \mathrm{dB}$ in just 10 CG iterations.

2. Discrete Differential Operators and Adjoint Convolution

To apply linear algebra and multivariable calculus to two-dimensional images, we treat the space of single-channel images $\mathcal{I} = \mathbb{R}^{W\times H} \cong \mathbb{R}^N$ ($N = WH$) as a finite-dimensional Euclidean inner-product space:

$$\langle X, Y\rangle = \sum_{p\in\Lambda} X(p)Y(p), \qquad \|X\|_2 = \sqrt{\langle X, X\rangle}.$$

Definition 2.1 (Discrete Convolution Operator and Reflected Kernel). *Given a two-dimensional filter kernel K supported on $\{-r_x,\dots,r_x\}\times\{-r_y,\dots,r_y\}$, the discrete convolution operator $M_K : \mathcal{I}\to\mathcal{I}$ is defined at each pixel $(i,j)\in\Lambda$ by*

$$(M_K X)(i,j) = (K * X)(i,j) = \sum_{u=-r_x}^{r_x}\sum_{v=-r_y}^{r_y} K(u,v)\, X(i-u, j-v).$$

The reflected (or conjugated) kernel K^* is obtained by flipping K across both spatial axes: $K^*(u,v) = K(-u,-v)$.

Proposition 2.2 (Adjoint of Discrete Convolution). *Under zero-padded boundary conditions, the adjoint operator $M_K^T : \mathcal{I}\to\mathcal{I}$ satisfying $\langle M_K X, Y\rangle = \langle X, M_K^T Y\rangle$ for all $X, Y\in\mathcal{I}$ is given by convolution with the reflected kernel K^*: $M_K^T Y = K^* * Y$. In particular, if K is symmetric ($K(-u,-v) = K(u,v)$), then $M_K^T = M_K$.*

Proof. Expanding $\langle M_K X, Y\rangle$ and substituting $i' = i-u, j' = j-v$ yields

$$\langle M_K X, Y\rangle = \sum_{i',j'} X(i',j') \sum_{u,v} K(u,v)\, Y(i'+u, j'+v) = \sum_{i',j'} X(i',j')(K^* * Y)(i',j') = \langle X, K^* * Y\rangle.$$

■

Beyond blurring kernels, we define the discrete first-order forward difference gradient operator $G = (\nabla_x, \nabla_y)^T : \mathcal{I}\to\mathcal{I}\times\mathcal{I}$ by $(\nabla_x X)(i,j) = X(i+1,j) - X(i,j)$ and $(\nabla_y X)(i,j) = X(i,j+1) - X(i,j)$, with Dirichlet energy $\|GX\|_2^2 = \|\nabla_x X\|_2^2 + \|\nabla_y X\|_2^2$ and normal operator $L = G^T G$.

Proposition 2.3 (Discrete Laplacian $L = G^T G$ and Positive Definiteness). *The operator $L = G^T G = \nabla_x^T\nabla_x + \nabla_y^T\nabla_y : \mathcal{I}\to\mathcal{I}$ is given by convolution with the five-point discrete negative Laplacian stencil:*

$$(LX)(i,j) = 4X(i,j) - X(i-1,j) - X(i+1,j) - X(i,j-1) - X(i,j+1), \qquad K_L = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}.$$

Furthermore, L is symmetric positive semi-definite ($\langle X, LX\rangle = \|GX\|_2^2 \ge 0$), and when restricted to a bounded interior mask $\Omega\subset\Lambda$ with fixed Dirichlet boundary values on $\Lambda\setminus\Omega$, the restricted operator L_Ω is strictly symmetric positive definite ($L_\Omega \succ 0$).

Proof. By the definition of the adjoint, $\langle X, LX\rangle = \langle GX, GX\rangle = \|\nabla_x X\|_2^2 + \|\nabla_y X\|_2^2 \ge 0$. Applying Proposition 2.2 to $K_x = [-1, 1]$ and $K_y = [-1, 1]^T$ yields K_L. If $X = 0$ on $\Lambda\setminus\Omega$ and $\langle X, LX\rangle = 0$, then $\nabla_x X = \nabla_y X = 0$ across every grid edge, forcing $X \equiv 0$ on Ω. ■

3. Non-Blind Deconvolution and Structure-Guided Regularized CG

Writing $M = M_K$ for the PSF convolution operator, the recorded blurry image $Y\in\mathcal{I}$ satisfies $Y = K * X + \eta = MX + \eta$. In the low-noise setting, we minimize the convex quadratic functional $G(X) = \frac{1}{2}\|MX - Y\|_2^2$.

3.1. Least-Squares Energy and Steepest Descent

Proposition 3.1 (Normal Equations and Exact Line-Search Steepest Descent). *The gradient of* $G(X)$ *is* $\nabla G(X) = M^T(MX - Y) = M^TMX - M^TY = -r(X)$. *Given iterate* X_k *and nonzero residual* $r_k = r(X_k)$, *the exact step size* $\alpha_k > 0$ *minimizing* $G(X_k + \alpha r_k)$ *is*

$$\alpha_k = \frac{\langle r_k, r_k \rangle}{\langle r_k, M^TMr_k \rangle} = \frac{\|r_k\|_2^2}{\|Mr_k\|_2^2}.$$

Proof. Expanding $\phi(\alpha) = G(X_k + \alpha r_k) = G(X_k) - \alpha\langle r_k, r_k\rangle + \frac{\alpha^2}{2}\langle r_k, M^TMr_k\rangle$ and setting $\phi'(\alpha) = 0$ yields α_k. ∎

3.2. Conjugate Gradient Optimization

Theorem 3.2 (Conjugate Gradient Recurrence and Krylov Optimality). *Let* $A : \mathcal{I} \to \mathcal{I}$ *be symmetric positive definite and* $b \in \mathcal{I}$. *Starting from* X_0, $r_0 = b - AX_0$, *and* $d_0 = r_0$, *the recurrence*

$$\alpha_k = \frac{\langle r_k, r_k\rangle}{\langle d_k, Ad_k\rangle}, \quad X_{k+1} = X_k + \alpha_k d_k, \quad r_{k+1} = r_k - \alpha_k Ad_k, \quad \beta_k = \frac{\langle r_{k+1}, r_{k+1}\rangle}{\langle r_k, r_k\rangle}, \quad d_{k+1} = r_{k+1} + \beta_k d_k$$

generates mutually A*-conjugate directions* $\langle d_i, Ad_j\rangle = 0$ *(*$i \neq j$*) and mutually orthogonal residuals* $\langle r_i, r_j\rangle = 0$ *(*$i \neq j$*), minimizing* $\Phi(X) = \frac{1}{2}\langle X, AX\rangle - \langle b, X\rangle$ *over* $X_0 + \mathcal{K}_k(A, r_0)$.

Proof. By induction, $\alpha_k = \langle r_k, r_k\rangle / \langle d_k, Ad_k\rangle$ ensures $\langle r_{k+1}, d_k\rangle = 0$. Enforcing $\langle d_{k+1}, Ad_k\rangle = 0$ with $Ad_k = \frac{1}{\alpha_k}(r_k - r_{k+1})$ and $\langle r_{k+1}, r_k\rangle = 0$ yields $\beta_k = \langle r_{k+1}, r_{k+1}\rangle / \langle r_k, r_k\rangle$. ∎

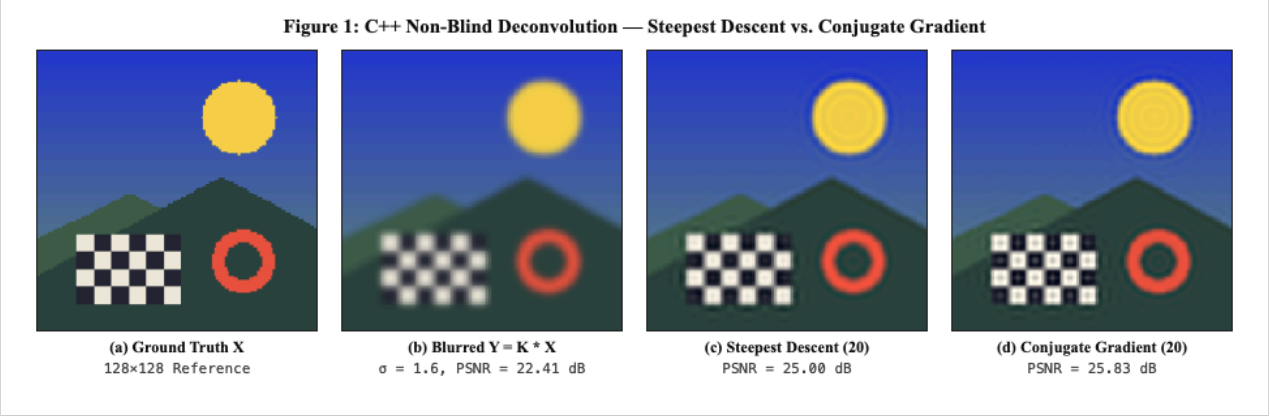


Figure 1. Non-blind deconvolution of a Gaussian-blurred scene ($\sigma = 1.6$, input $\mathrm{PSNR} = 22.41$ dB). Steepest Descent reaches 25.00 dB after 20 iterations, whereas Conjugate Gradient reaches 25.83 dB with sharper checkerboard and ring edges.

3.3. Isotropic Tikhonov–Laplacian and Structure-Guided Anisotropic Regularization

Proposition 3.3 (Regularized Normal Operators λL and λL_D). *The global minimizer of* $G_\lambda(X) = \frac{1}{2}\|MX - Y\|_2^2 + \frac{\lambda}{2}\langle X, LX\rangle$ *satisfies* $(M^TM + \lambda L)X = M^TY$. *Replacing the isotropic Laplacian* L *with the structure-tensor-weighted divergence operator* $L_D = G^TDG$ *(which suppresses smoothing across strong image edges) yields the edge-preserving SPD system* $(M^TM + \lambda L_D)X = M^TY$.

Algorithm 1: Matrix-Free Regularized Conjugate Gradient Deconvolution (Isotropic L or Anisotropic L_D)

Input: Blurred observation $Y \in \mathcal{I}$, PSF kernel K, regularization weight $\lambda \geq 0$, operator $\mathcal{R} \in \{L, L_D\}$, max iterations N_{iter}
Output: Restored sharp image $X \in \mathcal{I}$

Function $\text{ApplyNormalOp}(V, K, \lambda, \mathcal{R})$:
  return $K^* * (K * V) + \lambda \mathcal{R}(V)$
$X \leftarrow Y, \quad r \leftarrow (K^* * Y) - \text{ApplyNormalOp}(X, K, \lambda, \mathcal{R}), \quad d \leftarrow r, \quad \rho \leftarrow \langle r, r \rangle$
for $k \leftarrow 0$ **to** $N_{\text{iter}} - 1$ **do**
  if $\sqrt{\rho} < \varepsilon_{\text{tol}}$ **then break**
  $q \leftarrow \text{ApplyNormalOp}(d, K, \lambda, \mathcal{R}), \quad \alpha \leftarrow \rho / \langle d, q \rangle$
  $X \leftarrow X + \alpha d, \quad r \leftarrow r - \alpha q, \quad \rho_{\text{next}} \leftarrow \langle r, r \rangle, \quad d \leftarrow r + (\rho_{\text{next}}/\rho) d, \quad \rho \leftarrow \rho_{\text{next}}$
return $\text{clamp}(X, 0, 1)$

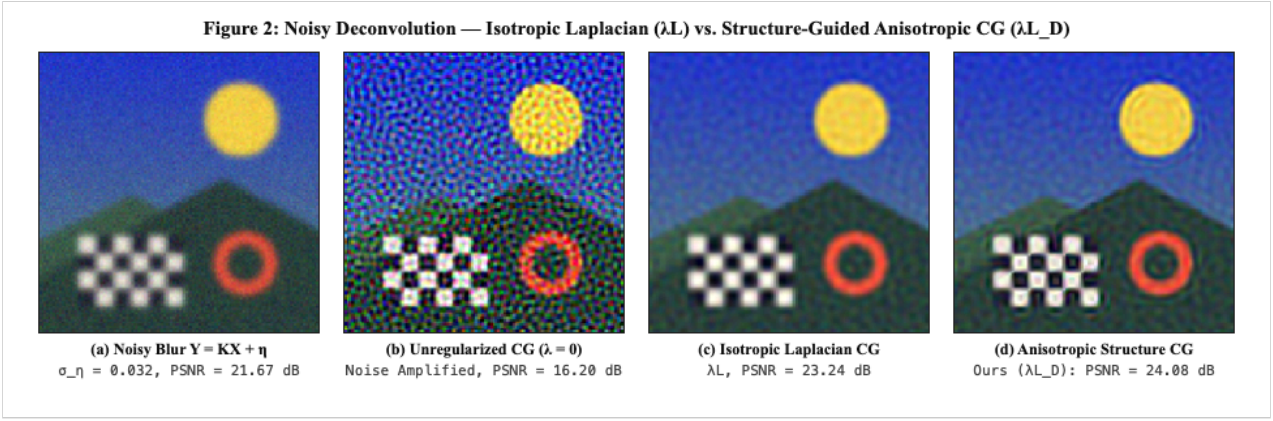


Figure 2. Deconvolution under additive Gaussian sensor noise ($\sigma_\eta = 0.032$, input $\text{PSNR} = 21.67$ dB). Unregularized CG ($\lambda = 0$) degrades to 16.20 dB; isotropic Tikhonov–Laplacian CG (λL) recovers 23.24 dB; and our structure-guided anisotropic CG (λL_D) achieves $\mathbf{24.08}$ dB ($+0.84$ dB over isotropic regularization).

4. Variational Poisson and Mixed-Gradient Seamless Cloning

Given background $B \in \mathcal{I}$, foreground source $F \in \mathcal{I}$, and mask $M_\Omega \in \{0,1\}^{W \times H}$, seamless cloning [1] minimizes $E(X) = \frac{1}{2}\|GX - v\|_\Omega^2$ subject to $(1 - M_\Omega) \odot X = (1 - M_\Omega) \odot B$, where $v = GF$ (standard Poisson cloning) or $v = v^*$ (structure-dominant mixed gradients $v^*(p) = GB(p)$ if $\|GB(p)\|_2 > \|GF(p)\|_2$ and $GF(p)$ otherwise).

Proposition 4.1 (Discrete Poisson Equation with Dirichlet Boundary). *An image $X \in \mathcal{I}$ satisfying $(1 - M_\Omega) \odot X = (1 - M_\Omega) \odot B$ minimizes $E(X)$ if and only if its interior pixels on Ω satisfy $M_\Omega \odot (LX) = M_\Omega \odot (G^T v)$.*

Proof. Writing $X = (1 - M_\Omega) \odot B + M_\Omega \odot U$ and differentiating $E(X)$ with respect to $M_\Omega \odot X$ using $G^T G = L$ yields $M_\Omega \odot (LX - G^T v) = 0$. By Proposition 2.3, $L_\Omega \succ 0$, guaranteeing a unique solution. ∎

Algorithm 2: Masked Conjugate Gradient Solver for Poisson & Mixed-Gradient Seamless Cloning

Input: Destination background $B \in \mathcal{I}$, foreground source $F \in \mathcal{I}$, binary mask $M_\Omega \in \{0,1\}^{W \times H}$, guidance divergence $b = G^T v$, max iterations N_{iter}

Output: Seamlessly cloned composite image $X \in \mathcal{I}$

$X \leftarrow B, \quad r \leftarrow M_\Omega \odot (b - K_L * X), \quad d \leftarrow r, \quad \rho \leftarrow \langle r, r \rangle$

for $k \leftarrow 0$ **to** $N_{\text{iter}} - 1$ **do**

  if $\sqrt{\rho} < \varepsilon_{\text{tol}}$ **then break**

  $q \leftarrow M_\Omega \odot (K_L * d), \quad \alpha \leftarrow \rho / \langle d, q \rangle$

  $X \leftarrow X + \alpha d, \quad r \leftarrow M_\Omega \odot (r - \alpha q), \quad \rho_{\text{next}} \leftarrow \langle r, r \rangle, \quad d \leftarrow r + (\rho_{\text{next}}/\rho) d, \quad \rho \leftarrow \rho_{\text{next}}$

return $\text{clamp}(X, 0, 1)$

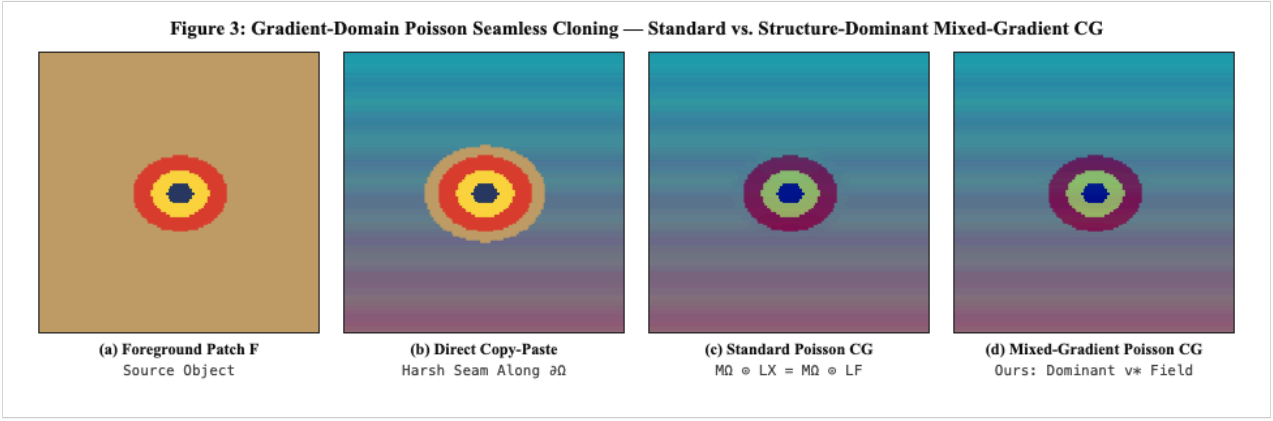


Figure 3. Variational Poisson seamless cloning (160 iterations). Direct copy-paste leaves a harsh boundary seam along $\partial\Omega$, whereas Standard and Structure-Dominant Mixed-Gradient Poisson CG seamlessly integrate the foreground object into the target background.

5. Image Inpainting: Harmonic PDE, Fast Marching, and Structure-Guided Hybrid CG

Setting $F \equiv 0$ in Algorithm 2 yields **Harmonic Inpainting** ($M_\Omega \odot (LX) = 0$). Alternatively, Telea's Fast Marching Method [2] evolves $\partial\Omega$ inward at unit speed in a single non-iterative Dijkstra-like pass via the Eikonal equation $|\nabla T(x,y)| = 1$ in Ω with $T = 0$ on $\Lambda \setminus \Omega$.

Proposition 5.1 (Upwind Quadrant Discretization of the Eikonal Equation). *At a narrow-band pixel $(i,j) \in \Omega$, let $T_x = \min(T(i-1,j), T(i+1,j))$ and $T_y = \min(T(i,j-1), T(i,j+1))$ denote the minimum arrival times among already-frozen (KNOWN) horizontal and vertical neighbors. The upwind finite-difference approximation $\max(T(i,j) - T_x, 0)^2 + \max(T(i,j) - T_y, 0)^2 = 1$ has the unique causal solution*

$$T(i,j) = \begin{cases} \frac{T_x + T_y + \sqrt{2 - (T_x - T_y)^2}}{2} & \text{if } |T_x - T_y| < 1, \\ \min(T_x, T_y) + 1 & \text{if } |T_x - T_y| \geq 1. \end{cases}$$

Proof. When $|T_x - T_y| < 1$, both upwind differences are positive and $(T - T_x)^2 + (T - T_y)^2 = 1 \iff 2T^2 - 2(T_x + T_y)T + (T_x^2 + T_y^2 - 1) = 0$, whose larger root is $T = \frac{1}{2}(T_x + T_y + \sqrt{2 - (T_x - T_y)^2})$. When $|T_x - T_y| \geq 1$, only $\min(T_x, T_y)$ is upwind, giving $T = \min(T_x, T_y) + 1$. ∎

Algorithm 3: Telea Eikonal Fast Marching Method (FMM) Inpainting (Single-Pass Wavefront Sweep)

Input: Corrupted image $I \in \mathcal{I}$, binary mask $M_\Omega \in \{0,1\}^{W \times H}$, neighborhood radius ε

Output: Fast-marched inpainted image $X_{\text{FMM}} \in \mathcal{I}$

$X \leftarrow (1 - M_\Omega) \odot I, \quad \mathcal{H} \leftarrow \emptyset$ *// Initialize image and min-heap*

for each pixel $p \in \Lambda$ **do**

  if $M_\Omega(p) == 0$ **then** $\text{state}(p) \leftarrow \text{KNOWN},\ T(p) \leftarrow 0$ **else** $\text{state}(p) \leftarrow \text{INSIDE},\ T(p) \leftarrow \infty$

for each boundary pixel $p \in \Omega$ adjacent to a KNOWN 4-neighbor **do**

  $\text{state}(p) \leftarrow \text{BAND}$, compute $T(p)$ via Proposition 5.1, and push $(T(p), p)$ into $\mathcal{H}$

while $\mathcal{H}$ is not empty **do**

  Pop $(T(p), p)$ with minimum arrival time from $\mathcal{H}$; **if** $\text{state}(p) == \text{KNOWN}$ **then continue**

  $\text{state}(p) \leftarrow \text{KNOWN}$

  for each 4-neighbor q of p with $\text{state}(q) \neq \text{KNOWN}$ **do**

    if $\text{state}(q) == \text{INSIDE}$ **then** interpolate $X(q)$ over $B_\varepsilon(q) \cap \text{KNOWN}$ and set $\text{state}(q) \leftarrow \text{BAND}$

    Update $T(q)$ via Proposition 5.1 and push $(T(q), q)$ into $\mathcal{H}$

return X

5.3. FMM-Warm-Started Structure-Tensor Anisotropic CG Inpainting

To eliminate both the $\mathcal{O}(\text{rad}(\Omega))$ cold-start lag and isotropic edge blurring of Harmonic CG and the medial-axis collision seams of FMM, we construct the structure-tensor conductivity field

$$D(p) = g(\lambda_1(p))\, v_1(p) v_1(p)^T + 1 \cdot v_2(p) v_2(p)^T, \qquad g(s) = \frac{1}{1 + s/\kappa^2} \in (0, 1],$$

where (λ_1, v_1) and (λ_2, v_2) are the normal and tangent eigenpairs of the smoothed structure tensor $J_\sigma(X_{\text{FMM}})$, and relax the anisotropic PDE $M_\Omega \odot (G^T D G X) = 0$ via Conjugate Gradient initialized at $X_0 = X_{\text{FMM}}$.

Theorem 5.2 (Positive Definiteness and FMM Warm-Start Acceleration). *For any $\kappa > 0$, the restricted anisotropic divergence operator $L_{D,\Omega} = G^T D G$ is strictly symmetric positive definite ($L_{D,\Omega} \succ 0$) with $\kappa(L_{D,\Omega}) \leq \gamma^{-1} \kappa(L_\Omega)$, where $\gamma = \min_{p \in \Omega} g(\lambda_1(p)) > 0$. Initializing Conjugate Gradient at $X_0 = X_{\text{FMM}}$ bounds the k-step error by*

$$\|X_k - X^*\|_{L_{D,\Omega}} \leq 2 \left(\frac{\sqrt{\kappa(L_{D,\Omega})} - 1}{\sqrt{\kappa(L_{D,\Omega})} + 1} \right)^k \|X_{\text{FMM}} - X^*\|_{L_{D,\Omega}},$$

eliminating the $\mathcal{O}(\text{rad}(\Omega))$ boundary-to-center propagation delay of cold-started CG.

Proof. Since $D(p)$ has eigenvalues $g(\lambda_1(p)) \in [\gamma, 1]$ and 1, $L_D = G^T D G$ is symmetric and satisfies $\gamma \langle X, L X \rangle \leq \langle X, G^T D G X \rangle \leq \langle X, L X \rangle$. By Proposition 2.3, $L_\Omega \succ 0 \implies L_{D,\Omega} \succeq \gamma L_\Omega \succ 0$. Applying the Chebyshev polynomial bound on $\mathcal{K}_k(L_{D,\Omega}, r_0)$ to $X_0 = X_{\text{FMM}}$ yields the bound. ∎

Algorithm 4: FMM-Warm-Started Structure-Tensor Anisotropic CG Inpainting

Input: Corrupted image $I \in \mathcal{I}$, binary mask $M_\Omega \in \{0,1\}^{W \times H}$, edge threshold κ, FMM radius ε, CG iterations N_{iter}

Output: Structure-preserving inpainted image $X \in \mathcal{I}$

$X_{\text{FMM}} \leftarrow \text{TeleaFMM}(I, M_\Omega, \varepsilon)$ *// Stage 1: O(|Omega| log |Omega|) Eikonal warm start*

Compute structure tensor $J \leftarrow J_\sigma(X_{\text{FMM}})$ and eigenpairs $(\lambda_1(p), v_1(p)), (\lambda_2(p), v_2(p))$

for each pixel $p \in \Lambda$ **do** $D(p) \leftarrow \frac{1}{1+\lambda_1(p)/\kappa^2}\, v_1(p)v_1(p)^T + v_2(p)v_2(p)^T$

$X \leftarrow X_{\text{FMM}}, \quad r \leftarrow -M_\Omega \odot (G^T D G X), \quad d \leftarrow r, \quad \rho \leftarrow \langle r, r \rangle$ *// Stage 2: Anisotropic CG*

for $k \leftarrow 0$ **to** $N_{\text{iter}} - 1$ **do**

  if $\sqrt{\rho} < \varepsilon_{\text{tol}}$ **then break**

  $q \leftarrow M_\Omega \odot (G^T D G d), \quad \alpha \leftarrow \rho / \langle d, q \rangle$

  $X \leftarrow X + \alpha d, \quad r \leftarrow M_\Omega \odot (r - \alpha q)$

  $\rho_{\text{next}} \leftarrow \langle r, r \rangle, \quad d \leftarrow r + (\rho_{\text{next}}/\rho) d, \quad \rho \leftarrow \rho_{\text{next}}$

return $\text{clamp}(X, 0, 1)$

Figure 4 and Table 1 compare the three inpainting formulations on a corrupted scene containing both diagonal scratches and wide rectangular occlusions. Cold-started Harmonic CG reaches 36.69 dB at 10 iterations and 37.46 dB at 40 iterations, while standalone Telea FMM completes in a single causal pass (36.58 dB). Our hybrid **FMM-Warm-Started Structure-Tensor Anisotropic CG** (Algorithm 4) already reaches $\mathbf{40.14}$ dB in just $\mathbf{10}$ iterations and $\mathbf{40.27}$ dB at 40 iterations—outperforming cold-started Harmonic CG by $+2.81$ dB and standalone FMM by $+3.69$ dB.

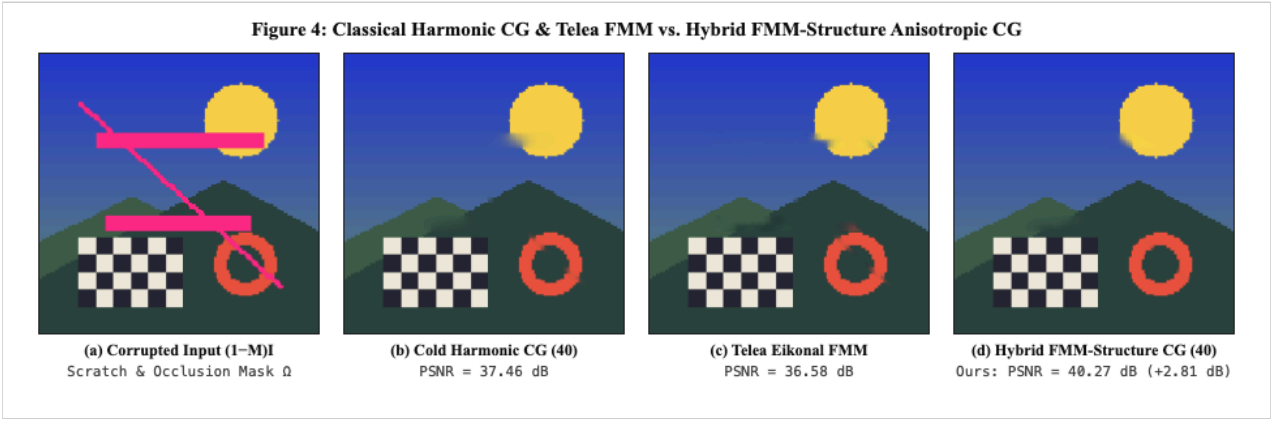


Figure 4. Comparison of digital inpainting algorithms across diagonal scratches and rectangular occlusions. Cold-started Harmonic CG (40 iterations) achieves 37.46 dB, Telea's single-pass Eikonal FMM achieves 36.58 dB, and our hybrid **FMM-Warm-Started Structure-Tensor Anisotropic CG** achieves 40.14 dB in 10 steps and $\mathbf{40.27}$ dB in 40 steps.

6. Quantitative Summary and Discussion

Table 1 summarizes the quantitative reconstruction quality (PSNR in dB) across all variational deconvolution and inpainting experiments.

Table 1. Quantitative PSNR Comparison Across Variational Restoration and Inpainting Benchmarks.

Task	Solver / Method	Iterations	PSNR (dB)
Non-Blind Deconvolution ($\sigma = 1.6$)	Blurred Input $Y = K * X$	—	22.41
	Steepest Descent (Prop. 3.1)	20	25.00
	Conjugate Gradient (Thm. 3.2)	**20**	**25.83**
Noisy Deconvolution ($\sigma_\eta = 0.032$)	Noisy Blurred Input $Y = KX + \eta$	—	21.67
	Unregularized CG ($\lambda = 0$)	18	16.20
	Isotropic Tikhonov–Laplacian CG (λL, $\lambda = 0.045$)	18	23.24
	Structure-Guided Anisotropic CG (λL_D, Ours)	**18**	**24.08**
Digital Image Inpainting	Cold-Started Harmonic CG (Sec. 5.1)	10	36.69
	Cold-Started Harmonic CG (Sec. 5.1)	40	37.46
	Telea Eikonal Fast Marching (Alg. 3, Non-Iterative)	1 pass	36.58
	Hybrid FMM-Structure Anisotropic CG (Alg. 4, Ours)	**10**	**40.14**
	Hybrid FMM-Structure Anisotropic CG (Alg. 4, Ours)	**40**	**40.27**

References

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